

An Intelligent Mobile Agent for Supporting Arabic-Speaking Blind Users

Background and Problem

Blind users rely on assistive technologies, such as screen readers, to interact with mobile applications. Although these tools are essential, completing multi-step tasks may require considerable effort to navigate interface elements, understand their content, and remember the sequence of actions needed to reach a goal.

These challenges may be greater in some Arabic applications due to differences in interface design and varying levels of accessibility support. This can affect users' ability to complete tasks independently or may require them to seek assistance from others.

Proposed Solution

This project proposes the development of an intelligent mobile agent that supports Arabic voice interaction and assists blind users in completing multi-step digital tasks.

The user expresses a goal through a natural voice command. The system then interprets the request, analyzes the current state of the mobile interface, and selects and executes the appropriate actions needed to achieve the goal. The agent also provides spoken feedback explaining what is happening and pauses to request the user's guidance or confirmation when necessary.

The system is intended to provide an intelligent assistance layer that complements existing accessibility tools. Its goal is to reduce the effort required to navigate applications while ensuring that the user remains in control of important actions and decisions.

Research Question

To what extent can a locally deployed, Arabic-enabled intelligent mobile agent accurately and safely interpret user goals and complete a selected set of multi-step Android tasks using accessibility information?

Project Scope

The project will focus on developing a prototype intelligent agent for the Android platform that supports interaction in Arabic. The system will be tested on a selected set of applications and multi-step tasks to investigate its ability to understand user goals, analyze interface states, select suitable actions, and execute them reliably.

The project aims to run the agent's core components locally on the mobile device wherever possible. This includes processing user commands, analyzing interface information, and making execution decisions. This on-device approach is intended to improve privacy and reduce dependence on cloud-based services.

The prototype is not intended to support every Android application. Instead, it will demonstrate the feasibility of the proposed concept within a clearly defined environment and scope. The project will also consider safety, privacy, and user control, particularly when actions are important or require explicit user approval.

Expected Outcomes

The project is expected to produce a functional prototype demonstrating the feasibility of using a local, Arabic-enabled intelligent agent to help blind users complete selected mobile tasks.

It is also expected to provide initial evidence regarding the agent's task-completion accuracy, reliability, safety, Arabic-language robustness, and on-device performance across the selected scenarios.

Project Value

The project addresses an important challenge in digital accessibility by combining artificial intelligence, Arabic voice interaction, computer-use technologies, and human-centered design.

It has the potential to create meaningful social impact by supporting the independence of Arabic-speaking blind users and improving their privacy through local on-device processing. The project may also contribute to applied research on intelligent agents, accessibility, and Arabic interaction with artificial intelligence systems.